

Glacial Outburst Flooding

Mathematical Geoscience

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1 Introduction

In this review, we aim to understand the phenomenon of glacial outburst flooding, also referred to as jökulhlaups. This occurs when a glacier containing a body of water, such as a subglacial lake or moraine-dammed lake suddenly yields and causes a release of the water, sometimes with devastating consequences for the surrounding region.

We may divide this phenomenon into multiple stages. The first stage is the accumulation of water, through surface ice melting, basal ice melting (the process of ice thinning at the base caused by the removal of ice due to melting and drainage), rainfall, or by geothermal heat and volcanic activity.

Next, the flood can be initiated by multiple different causes such as the collapse of the dam's structural integrity, by melting or buoyant forces of the subglacial water and erosion. Alternatively, this may be caused by the build up of hydraulic pressure in the water superseding the ice overburden pressure (the weight exerted by the overlying glacier ice on the water or sediments beneath it), or by volcanic activity or increases in geothermal heat, causing the ground level ice to melt rapidly.

What follows is a positive feedback loop whereby the flowing water widens the channels through which it flows, further accelerating its release and increasing discharge. This is a result of thermal erosion of the ice via the warmer water, abrasion of the ice boundary through sediment particles in the water and by the heat generated from the friction of the water shearing against the ice boundaries.

Finally, after peak discharge is reached and water pressure begins to subside, we see that through collapse or creep closure (the gradual deformation and inward movement of glacier ice due to its plastic flow under its own weight), the channels are resealed until the next outburst.

It is evident that jökulhlaups have a rich structure, and this has been the subject of much research, in particular around the mathematical models that attempt to understand the underlying processes of channel widening and narrowing, and how other physical factors affect the nature of individual jökulhlaup occurrences. Here, we shall focus predominantly on two contrasting models, namely the 'Nye model', developed by J. F. Nye in his 1976 paper [12], and the Spring-Hutter formulation, developed by U. Spring and K. Hutter from 1980 through 1982, though here we shall focus on the adaptation of G. K. C. Clarke in the model found in [5].

2 The Nye Model

The Nye model seeks to explain the physics of the jökulhlaup occurrences, focusing particularly on the interactions between the Grímsvötn volcano in Iceland and the Vatnajökull glacier that surrounds it.

2.1 Equations for the flow of water in tunnels and glaciers

We shall begin by deriving the time-dependent differential equations that govern the flow of water through a conduit in the glacier, where the conduit is being both melted by the friction resulting from the shearing of the water flow along its walls, and simultaneously contracted via plastic deformation as stated in [14]. Consider a water filled conduit on an incline as given in Figure 1 below, we find that the cross-

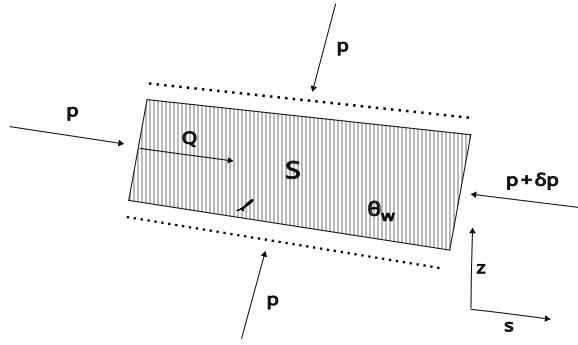


Figure 1: A schematic diagram of a section of conduit that allows us to define the governing equations of the Nye model of glacial outburst floods.

sectional area $S(s, t)$ is, as we had stated, governed by two processes, where s is the downstream co-ordinate, and t is time. Firstly, the melting rate, here denoted $m(s, t)$, i.e. the mass melted per unit length per unit time, and the change in $S(s, t)$ as a result of plastic deformation as a function of time, denoted $\left(\frac{\partial S}{\partial t}\right)_{pl}$. Combining, we find

$$\frac{\partial S}{\partial t} = v_i m + \left(\frac{\partial S}{\partial t}\right)_{pl}, \quad (1)$$

where v_i is the specific volume of ice, defined by $v_i = \rho_i^{-1}$, with ρ_i being the density of the ice. Defining $Q(s, t)$ as the flux of water flowing through cross-section $S(s, t)$, and $v_w = \rho_w^{-1}$, as the specific volume of water, where ρ_w is the density of water, we can derive the continuity equation applicable to our situation by noting

$$\frac{\partial}{\partial t} \int_{s_1}^{s_2} S \, ds := -Q \Big|_{s=s_1}^{s=s_2} = - \int_{s_1}^{s_2} \frac{\partial Q}{\partial s} \, ds,$$

leading to the expression

$$\frac{\partial}{\partial t} \int_{s_1}^{s_2} S \, ds = - \int_{s_1}^{s_2} \frac{\partial Q}{\partial s} \, ds + \int_{s_1}^{s_2} v_w m \, ds,$$

where it is clear the second term on the right-hand side of the equation is a source term, describing the mass of water added due to melting. Hence we recover a sourced continuity equation given by

$$\frac{\partial S}{\partial t} + \frac{\partial Q}{\partial s} = v_w m. \quad (2)$$

Using (1) and (2), we can find an explicit expression for the rate of plastic deformation, which yields the form

$$\left(\frac{\partial S}{\partial t} \right)_{pl} = - \frac{\partial Q}{\partial s} + m \Delta v, \quad (3)$$

where we have defined $\Delta v = v_w - v_i$ as the change in (specific) volume upon melting. We can also express plastic deformation using a reformulation of the Glen flow law, given by $\dot{\epsilon} = k \tilde{\sigma}^n$ [8], where $\dot{\epsilon}$ is the compressive strain rate, σ is the compressive stress, and n and k are positive constants. Reformulating in terms of the pressure and cross-sectional area and this allows us to write

$$\left(\frac{\partial S}{\partial t} \right)_{pl} = - \mathcal{K}_o S (p_i - p)^n, \quad (4)$$

where $\mathcal{K}_o$ is a constant depending on the shape of the cross-section (measured empirically), p_i is the ice overburden pressure and p is water pressure. We find that if S is circular, then $n \approx 3$ [11].

Next, we now wish to apply the first law of thermodynamics. Let us consider Figure 1 again and the length $l(s, t)$ of the conduit. As the conduit contracts by the external pressures around it, work is done by these surroundings. The rate of work done on the system along the assumed cylindrical boundary, denoted $\left(\frac{\partial W}{\partial t} \right)_{boundary}$, (via partial differentiation and re-expression of $W = \int \vec{F} \cdot d\vec{s}$) is given by

$$\left(\frac{\partial W}{\partial t} \right)_{boundary} = -pl \left(\frac{\partial S}{\partial t} \right)_{pl}. \quad (5)$$

Reformulating our work done expression once again, we may find that the rate of work done at the upper boundary is pQ , and using the definition of the product rule, the overall rate of work done across the conduit is

$$\left(\frac{\partial W}{\partial t} \right)_{conduit} = - \frac{\partial(pQ)}{\partial s} l. \quad (6)$$

The rate of work done by gravity is $Q\rho_w g_s l$ where g_s is the s-component of gravity, given by $g \sin \alpha$ where α is the angle between the vertical and the down-stream coordinate s . Therefore, by combining the above expressions, we find that the total rate of work done upon the conduit per unit length is given by

$$\left(\frac{\partial W}{\partial t}\right)_{total} = -\frac{\partial(pQ)}{\partial s} - p\left(\frac{\partial S}{\partial t}\right)_{pl} + Q\rho_w g_s. \quad (7)$$

As we are aiming for the rate of increase of internal energy in the system, we shall further assume some restrictions on the properties of the water in the conduit. Firstly, if the Reynolds number, $Re = \frac{\rho u L}{\mu}$, with ρ the density of the fluid, u the speed of the fluid, L the characteristic length of the system, and μ the dynamic viscosity of the fluid, is in the range $10^7 - 10^8$ [12], then we may safely assume that this flow is turbulent and further that the water is well mixed, at temperature $\theta_w(s, t)$. However, at the boundary between the water and the ice, the temperature of the water will fall to $\theta_i(s, t)$, the melting point of the ice. To help address this, we shall use a re-expression of the internal energy relation usually given by $U = Mc\Delta\theta$ (here, M denotes mass and c denotes a general specific heat capacity, with $\Delta\theta$ being a temperature gradient). Our system should contain two parts: firstly, the pre-existing water at time t and secondly, ice that has melted into water across timescale δt , along with the internal energy needed in order to melt the ice into water and raise it to the same temperature. The total sum of internal energy δU is hence the sum of the two terms stated,

$$\delta U = Sl\rho_w\sigma\frac{d\theta_w}{dt}\delta t + ml\delta t\{\Delta u + \sigma(\theta_w - \theta_i)\}, \quad (8)$$

where σ is the specific heat capacity of the water, Δu is the difference of internal energy per unit mass between the water and the ice at melting point, with the water assumed to be incompressible. Finally, $\frac{d}{dt}$ is the derivative following the element of the particle, otherwise referred to as the material derivative. We also neglect any contribution from kinetic energy because flow through conduits is assumed to be governed mainly by gravity and pressure differences. Here, Nye makes the assumption that if we neglect the conduction of heat across the ends of our moving system, the heat input of the system is zero. Hence, if the first law of thermodynamics is given by $\Delta U = \tilde{Q} + W$ ($\tilde{Q}$ here is heat added to the system), and we have assumed $\tilde{Q} = 0$, then we can rearrange and equate (7) and (8). We shall also use another reformulation of the first law by noting that W can be given by $-p\Delta v$, and $\tilde{Q}$ by L (with L the latent heat per unit mass), so we may re-cast Δu as $L - p\Delta v$, and so we can say

$$-\frac{\partial(pQ)}{\partial s} - p\left(\frac{\partial S}{\partial t}\right)_{pl} + Q\rho_w g_s = S\rho_w\sigma\frac{d\theta_w}{dt} + m\{L - p\Delta v + \sigma(\theta_w - \theta_i)\}, \quad (9)$$

again substituting the plastic deformation rate term by using (3) and recalling the product rule, we can simplify the expression to

$$Q \left(\rho_w g_s - \frac{\partial p}{\partial s} \right) - \rho_w \sigma S \frac{d\theta_w}{dt} = mL + m\sigma(\theta_w - \theta_i). \quad (10)$$

This equation has the interpretation that if the system is taken to be only the water entering volume Q without the addition made by melting, then we find $Q\rho_w g_s$ is the work done by gravity along the s direction. We also find $-Q\frac{\partial p}{\partial s}$ is the work done by the water pressure at the ends, and $\rho_w \sigma S \frac{d\theta_w}{dt}$ being the heat output used to raise the temperature of the water. The right-hand side of the expression is the heat output used in the change of state of the ice as well as the heating of it. Following [14], we employ the empirical Gauckler-Manning formula¹ for mean velocity of turbulent flow, $\bar{u} = \frac{Q}{S}$, finding the relation

$$\bar{u} = \frac{R^{\frac{2}{3}}}{n'} \left\{ \frac{1}{\rho_w g} \left(\rho_w g_s - \frac{\partial p}{\partial s} \right) \right\}^{\frac{1}{2}}, \quad (11)$$

where R is the hydraulic radius, given by the cross-section S divided by the perimeter of the cross section (in the case of a circular cross-section and semicircular cross-section, $R = \frac{r}{2}$ and $R = \frac{\pi r}{4+2\pi}$ respectively, where r is the radius). We also define n' as the Manning roughness coefficient, a positive constant. Recalling the definition of $\bar{u}$, we can reformulate (11) as

$$\rho_w g_s - \frac{\partial p}{\partial s} = \frac{\mathcal{N} Q^2}{S^{\frac{8}{3}}}, \quad (12)$$

where $\mathcal{N} = \left(\frac{S}{R^2}\right)^{\frac{2}{3}} \rho_w g n'^2$. We can see that the term $\frac{S}{R^2}$ depends only on shape, and not on size. Indeed, for a circular cross-section, we find this term is equal to 4π , for a semicircular cross-section, $\frac{8}{\pi} + 8 + 2\pi$. Note therefore that (12) holds as constant so long as $\mathcal{N}$ and n' are constant (meaning that neither the geometry nor the roughness of the problem do not vary significantly with scale).

When water at temperature θ_w flows turbulently down a tube of circular section whose walls are at temperature θ_i , the rate of the heat transfer to the walls is given by an empirical relation [10]:

$$Nu = 0.023 (Re)^{0.8} (Pr)^{0.4}, \quad (13)$$

¹Note, that Nye also refers to Williams, G. P. 1970. Manning formula - a misnomer? Journal of the Hydraulics Division, American Society of Civil Engineers, Vol. 96, and Mathews, W. H. 1973. Record of two jökulhlaups. Union Géodésique et Géophysique Internationale for this explicit expression, although I have not been able to find these sources.

with three dimensionless numbers, Nu , the Nusselt number, which can be thought of as the ratio between the heat transfer via convection and heat transfer via conduction in a fluid system, Re , the Reynolds number (which we shall re-express shortly) and Pr , the Prandtl number, which is a ratio of the momentum diffusivity and the thermal diffusivity of a fluid, and can be thought of as a ratio that describes how a fluid transfers and stores heat. These numbers are defined by

$$Nu = \frac{\bar{h}D}{\kappa} \quad Re = \frac{\bar{u}D\rho_w}{\eta} \quad Pr = \frac{\eta\sigma}{\kappa},$$

with $\bar{h}$ as the heat transfer coefficient, D as the diameter of the cross-section (assuming a circular tube), κ , the thermal conductivity of the fluid and η as the (dynamic) viscosity of the fluid. We find equation (13) fits observations for Re between 10^4 and 1.2×10^5 , Pr between 0.7 and 100, with $\frac{L_0}{D} < 60$, with L_0 being the length of the tube. We can also easily see that the heat rate transfer from the fluid to the walls per unit length of tube is given by

$$\bar{h}(\pi D)(\theta_w - \theta_i),$$

which we had previously expressed in (10) as

$$mL + m\sigma(\theta_w - \theta_i).$$

We note that Pr is defined solely by intrinsic material properties, and by direct calculation, using $\eta = 1.787 \times 10^{-3} \text{ Pa} \cdot \text{s}$, $\sigma = 4200 \text{ Jkg}^{-1}\text{K}^{-1}$, and $\kappa = 0.56 \text{ Wm}^{-1}\text{K}^{-1}$ (with measurements taken at 273.15 K), we find that $Pr = 13.5$. Hence, recalling the definition of Nu and using the relations above, we find

$$0.023\pi (Re)^{0.8}(13.5)^{0.4}\kappa(\theta_w - \theta_i) = mL + m\sigma(\theta_w - \theta_i), \quad (14)$$

with

$$Re = \frac{2Q\rho_w}{\pi^{\frac{1}{2}}S^{\frac{1}{2}}\eta}, \quad (15)$$

found by noting that $DS^{-\frac{1}{2}} = 2\pi^{-\frac{1}{2}}$. We are now in a position where we can collect the governing equations for unsteady flow through a conduit together. Firstly, the geometry and ice flow equation from combining (1) and (4):

$$\frac{\partial S}{\partial t} = v_i m - \mathcal{K}_o S(p_i - p)^n, \quad (p < p_i), \quad (16)$$

the continuity equation given by (2):

$$\frac{\partial S}{\partial t} = v_w m - \frac{\partial Q}{\partial s}, \quad (17)$$

the water flow equation from (12):

$$\rho_w g_s - \frac{\partial p}{\partial s} = \frac{\mathcal{N} Q^2}{S^{\frac{8}{3}}}, \quad (18)$$

the energy equation from (10):

$$Q \left(\rho_w g_s - \frac{\partial p}{\partial s} \right) - \rho_w \sigma S \frac{d\theta_w}{dt} = mL + m\sigma(\theta_w - \theta_i), \quad (19)$$

the heat transfer equation from (14) and (15):

$$0.205 \left(\frac{2Q\rho_w}{\pi^{\frac{1}{2}} S^{\frac{1}{2}} \eta} \right)^{0.8} \kappa(\theta_w - \theta_i) = mL + m\sigma(\theta_w - \theta_i). \quad (20)$$

In these equations, we take the course of the conduit, $\mathcal{N}$, along with the ice overburden pressure, $p_i(s, t)$, as given, leaving five unknown quantities (Q, S, p, m and θ_w , all functions of s and t) across our five equations (16) - (20) (θ_i is a function of p in accordance with the Clausius-Clapeyron relation). We also note that only (20) is derived on the assumption of the circular cross section of the conduit.

2.2 Applying the model to the Grímsvötn Jökulhlaup

We now turn to the particular situation of the Grímsvötn volcano and the frequent jökulhlaups that occur there, on average between every five to ten years [3]. We begin by arguing that the equations (16) - (20) are the valid equations that we should apply to this situation. Firstly, we assume that the jökulhlaups that occur around Grímsvötn flow through one or multiple tunnels rather than under the glacier in a sheet. This is because of the assumption $p < p_i$, for if this did not hold, we claim that the water would flood out of the conduit and lift the ice into a sheet, flowing underneath.

Next, we shall focus upon the governing equations, and simplify them accordingly. It is entirely plausible that the difference in temperatures between the water and the ice, $(\theta_w - \theta_i)$, is small, that is to say it is $\mathcal{O}(1)$. Therefore, nondimensionalising equations (19) and (20), we find $mL \gg m\sigma(\theta_w - \theta_i)$, because for water, we have $\frac{\sigma}{L} \ll 1$. Therefore, we neglect $m\sigma(\theta_w - \theta_i)$ in these equations.

Nye then makes the assumption that $\frac{d\theta_w}{dt} = 0$ in equation (19), i.e. that the material elements of water do not change in temperature as they flow through the tunnels from the glacial lake. This assumption is made for two reasons: firstly, if the water at the start of the flood is to be at 0.9°C and $Re = 1.4 \times 10^7$ through the tunnel, the water would emerge at the same temperature as it had entered the

conduit, similarly, at peak discharge, taking $\theta_w = 5.4^\circ C$ with $Re = 1.5 \times 10^8$, then the water would emerge from the other end of the tunnels at the same temperature. Secondly, Nye states it as a valid assumption because it leads to excellent agreement with the data and other observations. Recalling our derivation of (19), we see that setting this term to zero effectively assumes that the heat output used to raise the temperature of the water is negligible. Hence, we can now reduce our governing equations. Firstly, our geometry and ice flow equation remains unchanged:

$$\frac{\partial S}{\partial t} = v_i m - \mathcal{K}_o S (p_i - p)^n, \quad (p < p_i), \quad (21)$$

similarly for our continuity equation:

$$\frac{\partial S}{\partial t} = v_w m - \frac{\partial Q}{\partial s}, \quad (22)$$

and the water flow equation:

$$\rho_w g_s - \frac{\partial p}{\partial s} = \frac{\mathcal{N} Q^2}{S^{\frac{8}{3}}}. \quad (23)$$

Finally, the simplified energy equation:

$$Q \left(\rho_w g_s - \frac{\partial p}{\partial s} \right) = mL, \quad (24)$$

where we have neglected to re-express (20) as it now effectively only serves to calculate $(\theta_w - \theta_i)$, given by $\frac{mL}{0.205 (Re)^{0.8\kappa}}$. This relation may also be inaccurate since the recalling we had derived (20) using (14), which was derived numerically in [10] using $Re = 10^4$. However, as we had stated previously, our water flow through the conduit occurs approximately in the region $Re = 10^7 - 10^8$, and it remains the only equation with a geometry specific derivation, where we had assumed the cross-section of the conduit was circular.

Furthermore, in (23) and (24), both contain the term $\rho_w g_s - \frac{\partial p}{\partial s}$. We shall define the hydraulic potential ϕ , given by

$$\phi(s, t) = \rho_w g_s z - p(s, t), \quad (25)$$

where z the vertical height of the tunnel and is a function of s and t . It is clear therefore that $-\frac{\partial \phi}{\partial s}$ is a negative pressure gradient that drives water from the inlet to the outlet of the conduit. We can therefore see that

$$-\frac{\partial \phi}{\partial s} = -\rho_w g \frac{\partial z}{\partial s} - \frac{\partial p}{\partial s}.$$

Recalling our co-ordinates z and s , if we define a perpendicular bisector across z , intercepting s at an angle α , then one can easily see that $-\frac{\partial z}{\partial s} = \sin \alpha$, and hence by recalling the definition of g_s , we see that $-\frac{\partial \phi}{\partial s} = \rho_w g_s - \frac{\partial p}{\partial s}$.

Now, Nye further assumes that to model the growth of the jökulhlaup, while the tunnel is melting, the rate of plastic deformation of the ice in this instance is negligible, hence neglecting the final term of (21). Now applying this reduced expression to (22), we find that

$$\frac{\partial Q}{\partial s} = (v_w - v_i)m. \quad (26)$$

Using (21), (23), (24) and the expression for $-\frac{\partial \phi}{\partial s}$ in order to eliminate Q and m , we can see that

$$\frac{\partial S}{\partial t} = \mathcal{K}_1 S^{\frac{4}{3}}, \quad (27)$$

where we have defined

$$\mathcal{K}_1 = \frac{v_i}{L\mathcal{N}^{\frac{1}{2}}} \left(-\frac{\partial \phi}{\partial s} \right)^{\frac{1}{2}} S^{\frac{4}{3}}. \quad (28)$$

In general, as $\mathcal{K}_1$ is a function of $\mathcal{N}$ and $\frac{\partial \phi}{\partial s}$, it is a function of the geometry and shape of the tunnel, as well as the roughness, n' , along with s and t . However, Nye notes that the average value of $-\frac{\partial \phi}{\partial s}$ is simply the difference in hydraulic potential between the two ends of the conduit, given by $\rho_w g \frac{z_0}{l_0}$, where z_0 is the height of the lake water level above the tunnel outlet and l_0 is the total length of the tunnel. Nye states that the lake level z_0 only changes by around 8% throughout the jökulhlaup occurrence, and hence we shall treat this term as a constant. We shall continue with this argument by approximating each function in turn with its average value, allowing us to treat $\mathcal{K}_1$ as a constant. As a result, we may integrate (27) for an explicit expression for S :

$$S = \left(-\frac{3}{\mathcal{K}_1 t} \right)^3, \quad (29)$$

where $t = 0$ has been chosen in at the time when S becomes infinite. We shall proceed with the same argument by treating $\mathcal{N}$ and $\frac{\partial \phi}{\partial s}$ as constants, equal to their average values and by using (27) and (23) in order to obtain a relation between Q and S , allowing us to write a differential equation for Q :

$$\frac{dQ}{dt} = \mathcal{K}_2 Q^{\frac{5}{4}}, \quad (30)$$

where

$$\mathcal{K}_2 = \frac{4v_i}{3L\mathcal{N}^{\frac{3}{8}}} \left(-\frac{\partial \phi}{\partial s} \right)^{\frac{11}{8}}. \quad (31)$$

Integrating expression (30) gives us an explicit expression for Q , again where $t = 0$ is taken to be where Q becomes infinite, as in (29):

$$Q = \left(-\frac{4}{\mathcal{K}_2 t} \right)^4. \quad (32)$$

Nye refers to [13], which gives an accurate discharge hydrograph ($Q : t$ relation) of the jökulhlaup that occurred in Grímsvötn in 1972. Then, the discharge was summed and plotted and compared to equation (32). If $\mathcal{K}_2$ was taken to be $7.45 \times 10^7 \text{ m}^{\frac{3}{4}} \text{ s}^{\frac{3}{4}}$, this allows the plotted points of the hydrograph to be within $\pm 4\%$ of the theoretical expression for Q . Since we have estimated the value of $\mathcal{K}_2$, we can work out the average value of $\mathcal{N}$, which is equal to $709 \text{ m}^{-\frac{8}{3}} \text{ kg}$. This in turn, (again considering a circular cross-section) allows us to calculate the Manning roughness coefficient, corresponding to $n' = 0.12 \text{ m}^{-\frac{1}{3}} \text{ s}$. According to [14], n' has values ranging between 0.0089 and 0.1. Nye claims that his result is a consequence of there being multiple tunnels, not all of circular cross-section, and the fact we have neglected to model the movement of sediment and boulders.

We now turn to modelling how the jökulhlaup stops. In accordance with [1], the lake level is still 230 m above the level of the rock-glacier interface when the flood subsides. We now attempt to model the plastic deformation rate we had previously neglected. During the flood, the lake level falls by 100 m, our assumption is that this drop in water level causes a large increase in the factor of $(p_i - p)$, as a result we can no longer neglect it. By making the assumption that this pressure difference trebles, and by (4) and [11], we have that the rate of plastic deformation increases by a factor of 27, easily overcoming the rate of enlargement of the tunnel. The flux of water is at its maximum when these two forces are in equilibrium, and causes an abrupt stop of the flooding thereafter.

We note that the rate of plastic contraction increases suddenly during the peak of the flood. Nye states two reasons for this: firstly, the fall in lake level is not steady, (for example, Torricelli's law states that the velocity of an incompressible fluid is given by $v = \sqrt{2gh}$) by observing equation (32), we see that writing $\tau = -t$, such that τ is the time remaining before the asymptote of the expression, we get $Q \propto \tau^{-4}$. This means even with a constant lake cross-section, i.e. a lake with vertical sides such that each level has the same volume, we can infer that Q is proportional to the lake level:

$$z_\infty = z_0 - E_0 \tau^{-3}, \quad (33)$$

where E_0 is a constant, z_∞ is the starting value of z_0 at $\tau = \infty$. This relation can be found by noting that a uniform vertical cross section implies $-Q = \frac{dV}{dt} = A \frac{dz}{dt}$ where V is the volume of water and A is the surface area of the lake, and applying (32).

Secondly, because $p_i - p$ varies linearly with the lake level, we must have the form

$$p_i - p = P_\infty \left\{ 1 + \left(\frac{\tau_0}{\tau} \right)^3 \right\}, \quad (34)$$

where P_∞ and τ_0 are constants. Therefore, using (4), we can see that there is a proportionality relation between τ and the rate of plastic deformation:

$$\left(\frac{\partial S}{\partial t} \right)_{pl} \propto \left\{ 1 + \left(\frac{\tau_0}{\tau} \right)^3 \right\}^3. \quad (35)$$

The fact that the lake in reality has sloping sides makes this effect even more pronounced, indeed Nye states that the pressure difference trebling is a conservative assumption. This relation explains why it was plausible to neglect the effects of the deformation in the early parts of the outburst, and why the flux relation (32) is stopped after the point of maximum outflow, before the model predicts a singularity at $t = 0$.

3 The Spring-Hutter Model

In this section, we shall use an alternative framework to the Nye model, known as the Spring-Hutter model, in order to describe the jökulhlaups. This model is a more complete description of the velocity, temperature, discharge and cross-sectional area over the entire length of the conduit, making it more computationally challenging. This work shall resemble the analysis of Clarke in [5].

3.1 Motivations

The primary reasons for adopting an alternative model to the Nye model is because even though conditions presented in Grímsvötn allow for excellent agreement with the recorded data of the jökulhlaup events, there are other deductions and untested assumptions that later appeared non-physical. Such problems can point to more systematic failure of the understanding of the jökulhlaups which occur more broadly around the world, such as “Hazard lake” and Summit lake, both in Canada.

Clarke had previously used an adaptation of Nye’s model, where the jökulhlaup progression was assumed to be governed in the region of a ‘seal’, where the interplay of the melting of the ice walls of the conduit and the ‘creep closure’ would ultimately control the outburst [4]. Sometimes, such adaptations are referred to as ‘bottleneck models’. This theory was used in [2] to simulate some jökulhlaups in Iceland, though again, some assumptions were unsatisfactory.

3.2 Governing equations

The modified Spring-Hutter equations used by Clarke are stated below. Note, for continuity, we have used the same symbols were possible as in the Nye model. Below are the evolution equations for water mass, ice mass, linear momentum and internal energy respectively:

$$\frac{\partial p}{\partial t} = -\frac{1}{\beta S} \left\{ \frac{\partial S}{\partial t} + \frac{\partial(vS)}{\partial s} - \frac{m}{\rho_w} \right\}, \quad (36)$$

where β is the fluid compressibility, v is the cross-sectionally averaged fluid velocity,

$$\frac{\partial S}{\partial t} = \frac{m}{\rho_i} - 2 \operatorname{sgn}(p_e) \left\{ \frac{|p_e|}{nB} \right\}^n S, \quad (37)$$

where p_e is the effective pressure, given by $p_i - p$, n is the exponent of the Glen flow law as before, B is the constant coefficient of the Glen flow law when expressed as $(\frac{\dot{\sigma}}{B})^n$,

$$\frac{\partial v}{\partial t} = -\frac{\partial}{\partial s} \left(\frac{1}{2} v^2 + \frac{p}{\rho_w} + g z_k \right) - \frac{1}{\rho_w S} (m v + P_w \tau_0), \quad (38)$$

where z_k is the elevation of the conduit above the datum, P_w is the wetted perimeter of the conduit and $\tau_0 = \frac{1}{8} \langle f_R \rangle v |v|$ is the wall stress exerted by turbulent flow, with $\langle f_R \rangle = \frac{P_i f_i + P_b f_b}{P_i + P_b}$ the Darcy-Weisbach roughness (an alternative roughness expression to that of Manning) averaged over the conduit perimeter, with f_i and f_b being the roughness of ice walls and conduit bed respectively, and P_i and P_b being the ice wall perimeter and bed perimeter respectively. Finally,

$$\frac{\partial \theta_w}{\partial t} = -v \frac{\partial \theta_w}{\partial s} + \frac{1}{\rho_w \sigma S} \left\{ P_w \tau_0 v - m \left(L + \sigma(\theta_w - \theta_i) - \frac{v^2}{2} \right) \right\}. \quad (39)$$

with $\theta_i = c_T p$, with c_T being the pressure melting coefficient of ice. These equations were then solved using the numerical method of lines, where the equations are discretised in all but one variable, and then this semi-discrete problem is solved by integrating either the resulting ordinary differential equations (ODEs) or differential algebraic equations (DAEs).

Equations (36) and (37) imply that the tunnel is always saturated with water. Clearly, this cannot be the case for the final moments of a jökulhlaup in the event that the lake empties completely. However, as this occurs in a much shorter time frame, it is judged to be a sufficient approximation. Further analysing (37), we see that when p_e is negative, that is to say the water pressure is greater than the ice overburden pressure, so the area of conduit grows in time. Note that this assumes no water is able to seep through the ice into other drainage networks. Finally, this expression does not contain a term that considers the changing of the geometry of cross-sectional area S , only that its size fluctuates. Clarke states that the only justification for this is simplicity.

3.3 Results

We now turn to comparing the results from Grímsvötn jökulhlaups from the two methods. Firstly, we shall focus on the assumption $\frac{d\theta_w}{dt} = 0$ made in the Nye model. Using [2], Clarke plotted the observed flood hydrograph we see in Figure 2 from the 1986 Grímsvötn jökulhlaup, along with the computed hydrograph using the Spring-Hutter model.

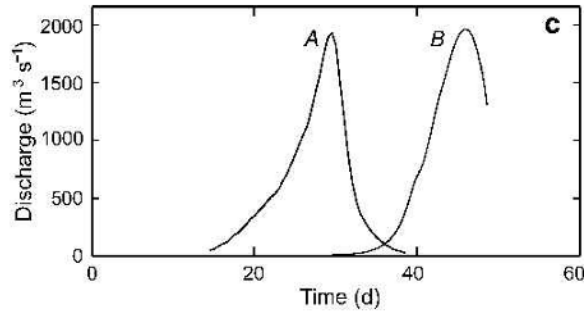


Figure 2: Image taken from [5]. We see a flood hydrograph from the 1986 Grímsvötn jökulhlaup, with A being the observed hydrograph taken from readings in [2], and B , the computed hydrograph from the Spring-Hutter model. Both curves used a lake temperature of 0.2°C .

The computed hydrograph used a semicircular conduit, with a prescribed perimeter averaged Manning roughness of $\langle n' \rangle = 0.032 \text{ m}^{-\frac{1}{3}}\text{s}$. To obtain a similar curve from the bottleneck model, Björnsson used a roughness of $n' = 0.08 \text{ m}^{-\frac{1}{3}}\text{s}$. This discrepancy of calculated Manning roughnesses was also found in similar hydrographs for “Hazard lake” and Summit lake. Although it can be partially explained by the

fact that the Spring-Hutter model was using a semicircular rock-floored conduit, even when the model is adapted to instead use a circular ice-walled conduit, a lower roughness is still required. The problem arises because the Nye postulate overestimates the rate of heat transfer to the walls, which in turn overestimates the cross-sectional area of the conduit and hence a high discharge rate, as shown in Appendix C of [5].

However, it should be noted that some refute the Manning roughness discrepancy of the Nye model as an inherent problem to the model itself. Fowler [7] was able to show that Nye’s model could yield a Manning roughness coefficient of $n' = 0.04 \text{ m}^{-\frac{1}{3}} \text{ s}$, which enabled a quantitative fit to the 1972 hydrograph, as seen in Figure 3. Rather than assuming terms constant to their average value, Fowler instead solved a ‘reduced model’ of Nye’s equations through the nondimensionalisation of terms, allowing him to solve a system of fewer equations via a numerical method explained in [6]. Figure 3 shows six recorded discharge hydrographs of Grímsvötn jökulhlaups from 1922, to 1986, with a normalised outputted numerical solution he had obtained.

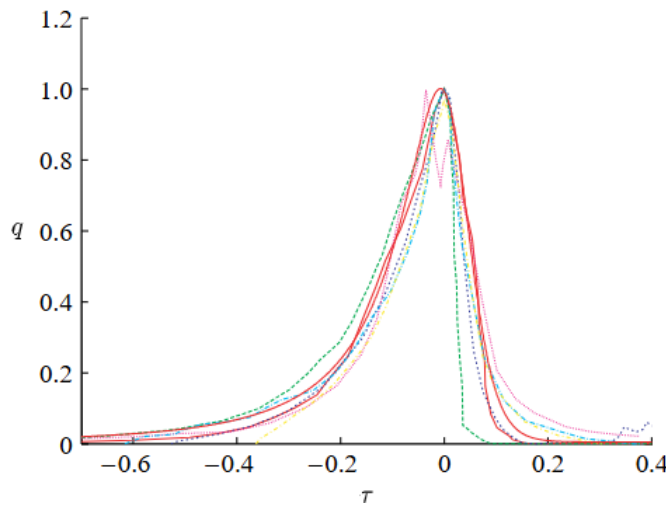


Figure 3: Taken from [7]. Six normalised hydrographs from 1922 (red), 1934 (green), 1954 (cyan), 1972 (magenta), 1976 (yellow) and 1986 (blue), with the numerical solution of the reduced model overlaid in red. In Fowler’s model, q is a scaling of the nondimensional flux, denoted Q^* . Similarly for $\tau \propto t^*$, where t^* is a nondimensional variable denoting time.

From Figure 3, we see that all Grímsvötn jökulhlaups have similar hydrograph profiles, which are in good agreement with the numerical solution of the reduced model. The main difference between each of the normalised hydrographs, and the model solution is that the termination of each jökulhlaup occurs across different timescales.

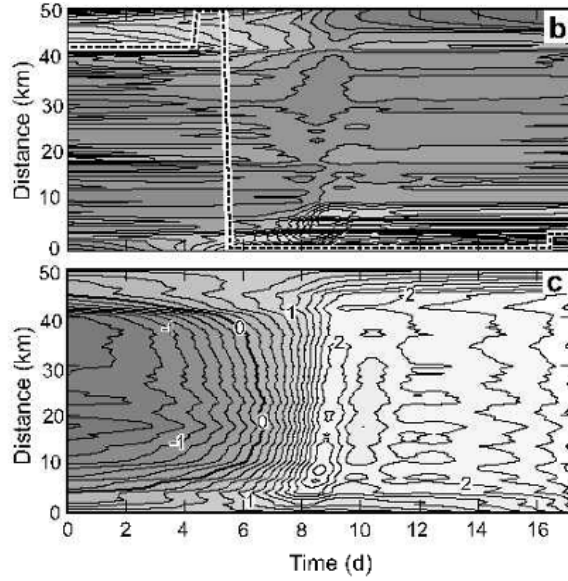


Figure 4: Two plots from [5], the top displaying the spatio-temporal evolution of $-\frac{\partial\phi}{\partial s}$ of a simulated Grímsvötn jökulhlaup. The darker shading corresponds to regions where energy is dissipated, and hence where the flow is restricted. The bottom, a spatio-temporal map of the effective pressure, p_e across this simulated outburst, with some annotated contours.

We see that the 1934 jökulhlaup terminates at around $\tau = 0.08$, whereas the 1972 jökulhlaup asymptotes to the axis at around $\tau = 0.4$. One plausible explanation is that the 1934 jökulhlaup was triggered by a volcanic eruption of Grímsvötn, and hence the resulting outburst was of immense magnitude [9].

Next, we shall turn to two more assumptions made by Nye: firstly, that the hydraulic potential gradient, $\frac{\partial\phi}{\partial s}$, can be approximated as the average hydraulic potential gradient, denoted $\langle \frac{\partial\phi}{\partial s} \rangle$, and secondly, that the water pressure p never exceeds the ice pressure p_i . In Clarke’s formulation, this is identical to stating that $p_e > 0$. In order to test the first assumption, we shall write $\phi = \rho_w g z_k + p$ and look for extrema in the values of $-\frac{\partial\phi}{\partial s}$. Let us denote $s^*(t)$ as the position where $\frac{\partial\phi}{\partial s}$ attains a minimum, or where $-\frac{\partial\phi}{\partial s}$ attains a maximum. In order for the Nye assumption to be valid, it is clear that the $-\frac{\partial\phi}{\partial s}$ should vary with time but not in down-flow distance.

It is clear from Figure 4 that there is a large number of contours in the hydraulic potential plot with increasing distance. This indicates that in the case of the Grímsvötn jökulhlaup, the Nye assumption fails. Furthermore, we see from the lower plot that Nye’s assumption that water pressure always remained less than ice

overpressure p_i , was also untrue. Indeed, one sees that for the majority of the flow in the early stages of the jökulhlaup, this is an incorrect assumption. It remains so until around $t = 6.5$ days. We can also assume that the complexity of these gradient plots arises from the complex topology used in the simulation.

We now turn to the issue of temperature of the water throughout the jökulhlaup, where both models encounter difficulties for numerous reasons. We are given from [2] that the temperature of the glacial lake in Grímsvötn is 0.2°C , and we shall now see from Figure 5 how the flood hydrographs are impacted by differences in water temperature.

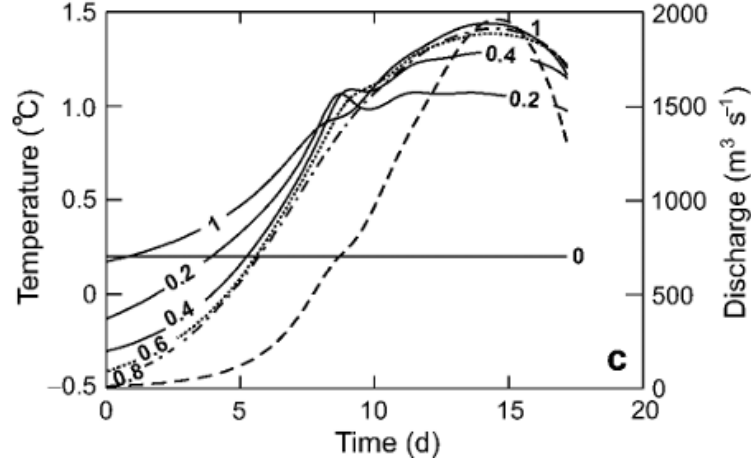


Figure 5: A spatio-temporal time evolution simulation of the temperature and discharge of a Grímsvötn jökulhlaup at different lengths, with the simulated flood hydrograph as dashed lines overlaid. Labelled curves correspond to the ratio of the distance from inlet to outflow in terms of conduit length l_0 , where 0 is the conduit inlet and 1 is the conduit outlet. Taken from [5].

We see from Figure 5 that the lake temperature is only slightly above the freezing temperature, and the flood is enabled through viscous dissipation, where work is done by fluid layers shearing causing heat. By $t = 6$ days, the temperature of the water is everywhere above the temperature of the lake, as the flood begins to enter its rapid growth (as seen by the increase of the gradient of the lines). The simulated maximum temperature at the outlet is around 1.4°C , in contrast to the much lower lake temperature. Crucially, such observations have not been made in reality, pointing to a failing of the Spring-Hutter model.

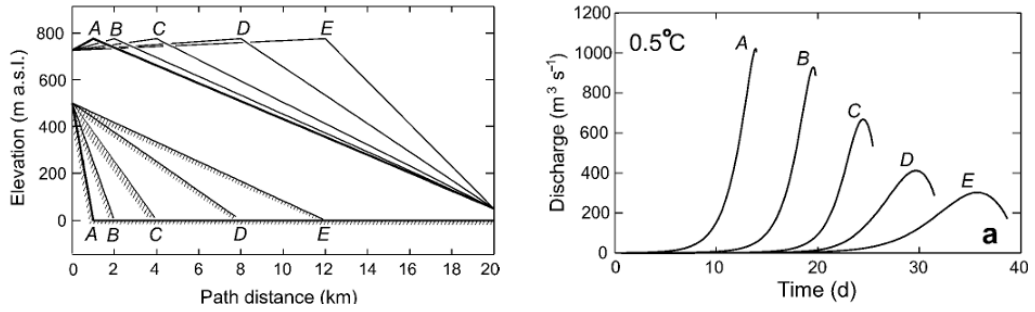


Figure 6: Taken from [5]. (Left) Five simple glacial topographies with identical values for inlet and outlet elevations, as well as the ice thickness at the seal. Maximum surface elevation and minimum bed elevation are labelled *A* to *E*, with *A* having the seal closest to the lake, and *E* furthest downstream. (Right) The effects of water temperature and geometry of the seal and glacier on the simulations of the discharge hydrographs using the bottleneck model, with a lake temperature of 0.5°C . At low temperatures, this discrepancy is greatest compared to higher lake temperatures, and hence presents a challenge for this method in the modelling of Grímsvötn jökulhlaups.

This problem becomes especially pronounced if such data is prescribed and used to simulate flood hydrographs using the bottleneck models adapted from the Nye model. Indeed, we see that the topography of the glacier and hence the position of the seal has significant impact on peak discharge, as seen in Figure 6.

Bottleneck models require four inputs: $z_k(0)$, the input elevation, $z_k(l_0)$, the output elevation, $z_k(s_{seal})$, the location of the seal of the conduit, and $z_i(s_{seal})$, the ice surface elevation at the seal. Clarke used the position of s_{seal} to vary from 1 km to 12 km from *A* to *E* respectively, a conduit length of 20 km and for each geometry we have $z_k(0) = 500$ metres above sea level (m.a.s.l.), $z_k(l_0) = 0$ m.a.s.l., $z_k(s_{seal}) = 0$ m.a.s.l. and $z_i(s_{seal}) = 800$ m.a.s.l.. The model used a semicircular conduit with a perimeter averaged Manning roughness of $\langle n' \rangle = 0.04 \text{ m}^{-\frac{1}{3}} \text{ s}$ for each geometry.

4 Conclusions

The Nye and the Spring-Hutter models have been instrumental in expanding our understanding of the phenomena of jökulhlaups, most notably, both models have provided accurate computed flood hydrographs as shown in Figures 2 and 3, though we have noted some weaknesses and shortcomings found in both. In [5], Clarke states that more intricate models or adaptations to existing models are hampered by the lack of physical data of lakes and glaciers before, during and after jökulhlaups. He

states that the temperature of the lake and the outburst flood data are rarely taken together, which we have seen in our simulation can have large impacts upon the shape and magnitude of the flood hydrographs, as well as on the resulting hydraulic roughness calculations. For example, some of our discussions surrounding Grímsvötn had been in relation to the 1986 flood, though Clarke states that the majority of the data collected was obtained in 1991, and applied retrospectively to that jökulhlaup. We know that this is a significant barrier to progress; from [2], we can see that lakes clearly do not exhibit isothermal properties as our models had implicitly assumed. Also according to [2], the heat transfer from the water during jökulhlaups may be greater than that considered by the Spring-Hutter model, since at high Reynolds numbers, the empirical heat transfer equations may be inaccurate. Such sentiment is reflected by Fowler in [7], where he states that heat transfer for $(Re) \gtrsim 10^8$ may be more efficient than that suggested by (13). Clarke also states that the conventional heat transfer equations used may not strictly apply to our situation, where a primary source of heat transfer was as a result of viscous dissipations between the fluid and the friction of those fluid layers against the walls of the conduit. Another conceivable explanation by Björnsson, is since the Reynolds number is so high, a lower temperature of the outflow water could be as a result of ice crystals and cooler sediment suspended in the water that have been eroded off the walls of the conduit through turbulence.

We also see that unlike the assumption made in the Nye model, the water pressure can exceed the ice overburden pressure for a significant period of the Grímsvötn jökulhlaup phenomena. The Spring-Hutter model is also not free of criticism in this regard, since from observations, if the water pressure significantly exceeds the ice overburden pressure, the water may create englacial cracks or flow into other drainage pathways, as water exchange between the conduit and surroundings is also not considered. Finally, we had showed for the Nye model, that the average of the hydraulic potential gradient $\langle \frac{\partial \phi}{\partial s} \rangle$ was not a good estimate of the potential gradient throughout the flow path. Nor was it an accurate the adaptation of the bottleneck model that the ‘seal’ should stay fixed in one location, as Figure 4 showed us that the position of the ‘seal’, given by the darkest shading, can move throughout the flooding.

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